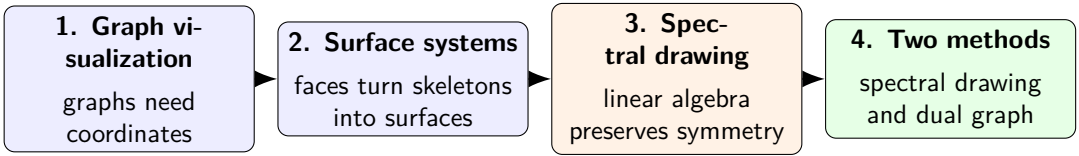
Constructing Surfaces from Cayley Graphs

Spectral Drawing and Face Systems

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Roadmap



Guiding question

A Cayley graph gives a highly symmetric **skeleton**. How do we find a natural **surface system** on it?

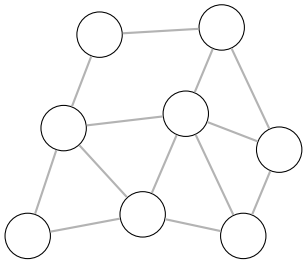
Graphs: One Language for Many Disciplines

A graph has only two ingredients

- **Vertices**: objects
- **Edges**: relationships between them

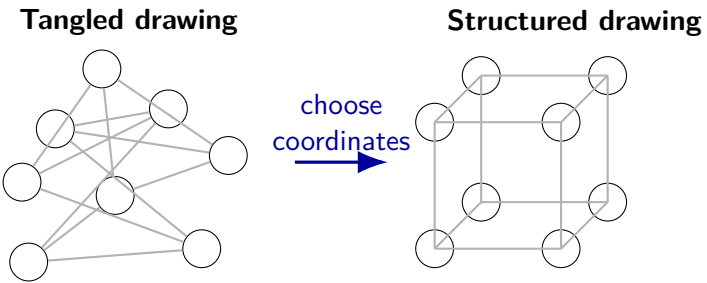
The same language appears everywhere:

- Social networks (people, friendships)
- Molecules (atoms, bonds)
- Maps and road networks
- Communication networks
- **Algebra and symmetry** ← our focus



Question: what hidden shape is this graph trying to show us?

Why Visualize Graphs?



Main idea

A graph is abstract data. A drawing is an interpretation. Good coordinates can reveal hidden symmetry and geometry.

Cayley Graphs: Graphs Built from Symmetry

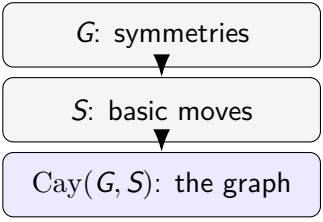
Idea

Start with a group G – a collection of symmetries – and a set S of basic moves.

- Every symmetry becomes a **vertex**.
- From each vertex g, draw an edge to gs for every basic move s.

Result: a highly symmetric graph where every vertex looks identical.

Because of this uniformity, the graph inherits the symmetry of G.



Key property

Cayley graphs are vertex-transitive: every vertex has the same local view.

Running Example: D_4, the Symmetries of a Square

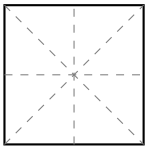
D_4 has 8 symmetries

- 4 rotations: $0°, 90°, 180°, 270°$
- 4 reflections: across 4 axes

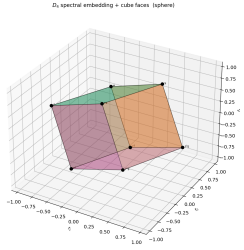
Choosing 3 basic moves gives a 3-regular Cayley graph:

$$V = 8, \qquad E = \frac{8 \cdot 3}{2} = 12.$$

Think of this graph as a **wireframe**: vertices and edges are present, but there is *no surface yet*.

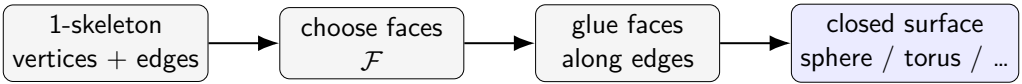


A square has 8 symmetries



A drawing of the D_4 Cayley graph

From a Graph to a Surface



Face system

A face system $\mathcal{F}$ is a choice of cycles to fill as faces, with the edges glued consistently to form a closed surface.

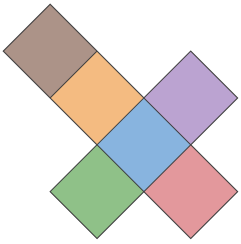
Euler characteristic

$$\chi = V - E + F$$

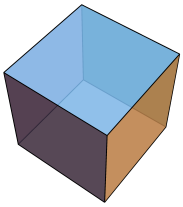
Closed orientable surfaces

$$\chi = 2 - 2g$$

The Key Example: D_4 Face System A



Fundamental domain / net



3D surface realization

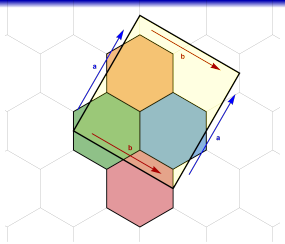
Cube / sphere system

The same D_4 skeleton can be filled by six square faces.

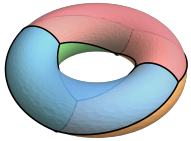
$$V = 8, \qquad E = 12, \qquad F = 6, \qquad \chi = 8 - 12 + 6 = 2.$$

Hence the resulting closed orientable surface is S^2.

The Key Example: D_4 Face System B



Fundamental domain



3D torus surface

Hexagon / torus system

The same D_4 skeleton can also be filled by four hexagonal faces.

$$V = 8, \qquad E = 12, \qquad F = 4, \qquad \chi = 8 - 12 + 4 = 0.$$

Hence the resulting closed orientable surface is T^2.

Purpose of the D_4 example

The skeleton is identical. The **face system** changes the topology.

What Changes? What Does Not?

	System A (sphere)	System B (torus)
Underlying graph	same D_4	same D_4
Vertices	8	8
Edges	12	12
Faces	6 squares	4 hexagons
Euler characteristic	$+2$	0
Surface	sphere S^2	torus T^2

The central tension

The graph alone cannot determine the surface type. We need extra data: the face system $\mathcal{F}$.

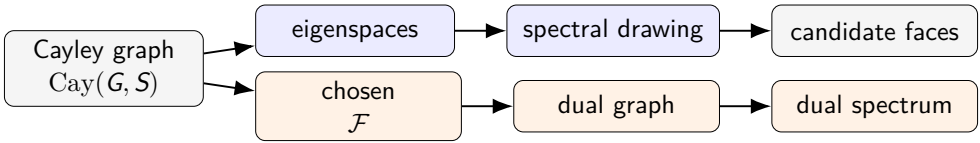
Two Ways to Study Surface Systems

Method 1: spectral drawing

- Start from the Cayley graph.
- Use eigenspaces as coordinates.
- Let the geometry suggest possible faces.

Method 2: dual graph

- Start from a chosen face system $\mathcal{F}$.
- Put one dual vertex in each face.
- Use face adjacency to study the surface.



Important distinction

Spectral drawing **suggests** possible geometry; the dual graph **describes** a chosen face system.

Spectral Drawing: A Linear Algebra Visualization

From graph to matrix

Build the adjacency matrix A:

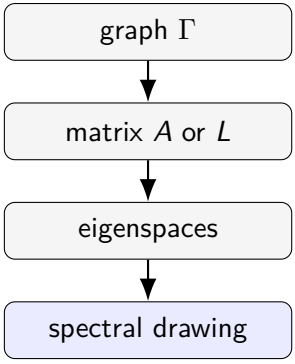
$$A_{uv} = 1 \quad \text{if vertices } u, v \text{ share an edge.}$$

For a regular graph, the Laplacian $L = dI - A$ has the same eigenspaces as A.

Coordinates from eigenspaces

Choose an eigenspace with basis $\psi_1, \ldots, \psi_k$ and draw

$$g \longmapsto \big(\psi_1(g), \ldots, \psi_k(g)\big) \in \mathbb{R}^k.$$



Eigenvalues alone are numerical.
Eigenspaces give coordinate functions.

Why Spectral Drawing Respects Symmetry

Cayley graph symmetry

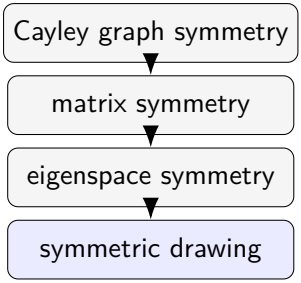
The group G acts on the vertices of $\mathrm{Cay}(G, S)$ by left multiplication. This action preserves adjacency.

Linear algebra consequence

Because the adjacency matrix respects the group action, its eigenspaces also respect the group symmetry.

Use the whole eigenspace

A basis can rotate, but the eigenspace itself is intrinsic. Therefore the drawing is well-defined up to an orthogonal change of coordinates.



D_4: A 3-Dimensional Eigenspace Suggests a Sphere

Spectral drawing

Use a 3-dimensional eigenspace as coordinates:

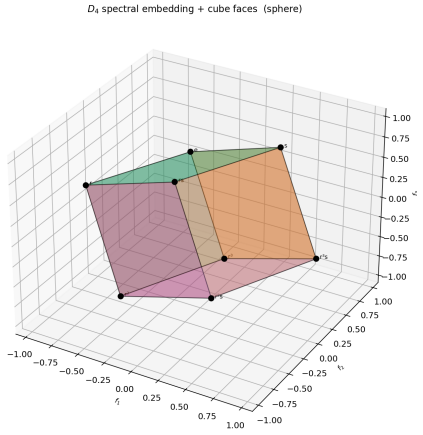
$$\phi(g) = \big(\psi_1(g), \psi_2(g), \psi_3(g)\big) \in \mathbb{R}^3.$$

For the D_4 Cayley graph, the vertices arrange into a cube-like shape.

Geometric suggestion

The cube-like drawing suggests the six square faces, hence a spherical face system:

$$V = 8, \quad E = 12, \quad F = 6, \quad \chi = 2.$$



The symmetry emerges from the eigenspace.

Spectral Drawing vs. Surface Systems

What spectral drawing can do

- Give symmetry-preserving coordinates.
- Reveal a natural geometric realization.
- Suggest candidate cycles to fill as faces.

What it does not decide

- It does not contain the face system as data.
- The same graph may still admit another surface system.
- The candidate must be verified combinatorially.

D_4 lesson

The 3D spectral drawing explains why the cube/sphere system is natural. It does **not** explain the torus system by a $2+2$ eigenspace pattern; the torus system is a different combinatorial choice of faces.

$\mathbb{Z}_3 \times \mathbb{Z}_3$: A $2+2$ Pattern Suggests a Torus

The Cayley graph

Take

$$G = \mathbb{Z}_3 \times \mathbb{Z}_3, \qquad S = \{\pm e_1, \pm e_2\}.$$

This is the 3×3 periodic grid.

$2+2$ Fourier coordinates

The two cyclic directions give two real 2-dimensional coordinate systems:

$$\mathbb{R}^2 \oplus \mathbb{R}^2.$$

Geometrically this suggests

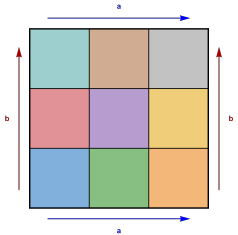
$$S^1 \times S^1,$$

which is a torus.

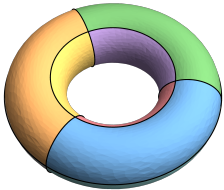
Technical note

For the ordinary unweighted adjacency matrix, the two 2D pieces may appear inside one 4D eigenspace. The natural splitting comes from the two Fourier directions.

$\mathbb{Z}_3 \times \mathbb{Z}_3$: Square Torus Surface System



Fundamental domain



3D torus drawing of the surface graph

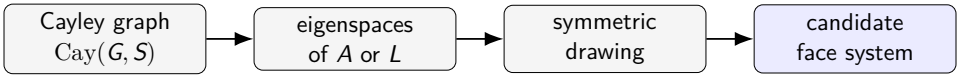
Fill by squares

The 3×3 periodic grid has 9 square faces:

$$V = 9, \qquad E = 18, \qquad F = 9, \qquad \chi = 9 - 18 + 9 = 0.$$

So the resulting closed orientable surface is a torus T^2.

Method 1: From Spectral Drawing



Spherical signal

A natural 3-dimensional eigenspace can place vertices in $\mathbb{R}^3$ and suggest a spherical polyhedral system.

$$3D \quad \rightsquigarrow \quad S^2\text{-type geometry}$$

Toroidal signal

A natural $2+2$ splitting gives two circular directions and suggests a torus.

$$2D + 2D \quad \rightsquigarrow \quad S^1 \times S^1$$

Verification step

Spectral drawing suggests; the face system and Euler characteristic verify.

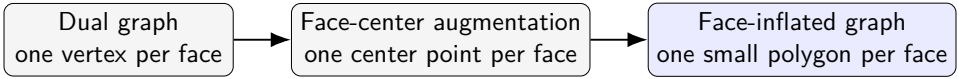
Method 2: Encoding the Face System

Problem with the ordinary dual graph

Given a graph with a chosen face system $\mathcal{F}$, the dual graph Γ^* has

- one vertex for each face,
- one edge when two faces share a boundary edge.

The dual graph records face adjacency, but it forgets the original Cayley vertices and the cyclic order around each face.



New goal

We want an augmented graph that remembers the face system while still preserving the symmetry of the original Cayley graph.

First Attempt: Face-Center Augmentation

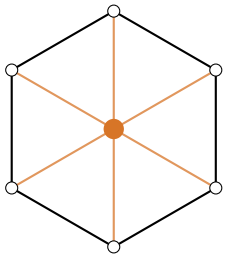
Construction

For each face $F \in \mathcal{F}$, add one new vertex c_F. Then connect c_F to all boundary vertices of F:

$$v \sim c_F \iff v \in \partial F.$$

What it preserves

- the original Cayley graph skeleton,
- which vertices lie on which face,
- the group symmetry, if $\mathcal{F}$ is G-invariant.



one face replaced by one center point

Limitation

It collapses the whole face to one point, so it loses the cyclic order around the face.

Better Attempt: Face-Inflated Graph

Construction

For a face

$$F = (v_1, v_2, \ldots, v_k),$$

add a smaller inner k-gon

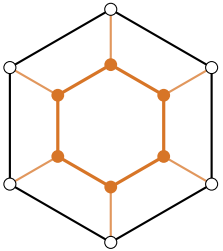
$$u_{F,1}, u_{F,2}, \ldots, u_{F,k}.$$

Add inner polygon edges

$$u_{F,i} \sim u_{F,i+1},$$

and radial edges

$$v_i \sim u_{F,i}.$$

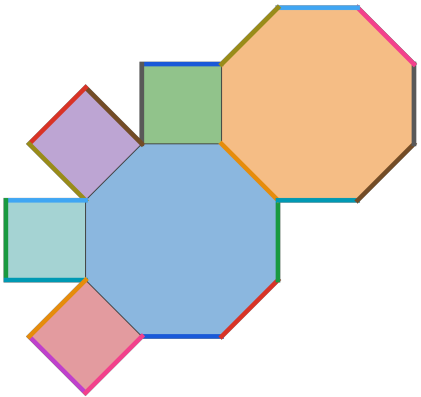


one face replaced by a small polygon

Role

This graph is a better object for applying spectral drawing to a chosen surface system.

Q_8: A Higher-Genus Cayley Surface



A genus-two face system on a Cayley graph of Q_8

Why include this example?

This example shows that the construction is not limited to sphere and torus cases.

Euler characteristic

$$V = 8, \qquad E = 16, \qquad F = 6.$$

$$\chi = 8 - 16 + 6 = -2.$$

Thus

$$2 - 2g = -2, \qquad g = 2.$$

What Each Example Does

Example	**Purpose**
D_4 with two face systems	Shows that the same Cayley graph can support different surfaces.
D_4 spectral cube	Shows how a 3-dimensional eigenspace suggests a spherical face system.
$\mathbb{Z}_3 \times \mathbb{Z}_3$ periodic grid	Gives the clean $2 + 2$ Fourier-coordinate example suggesting a torus.
Q_8 genus-two surface	Shows higher-genus surface systems; symmetric spectral drawing remains open.

What This Project Shows

1. Cayley graphs give highly symmetric graph skeletons built from group theory.
2. A surface requires extra information: the **face system** $\mathcal{F}$.
3. The same Cayley graph can support topologically different surfaces.
4. **Spectral drawing** uses eigenspaces to give symmetry-preserving coordinates.
5. A 3D eigenspace can suggest a spherical system; a $2 + 2$ structure can suggest a toroidal system.
6. The **dual graph** records face adjacency after a face system is chosen.

Main takeaway:

Spectral drawing suggests; face systems verify; dual graphs record.

Summary

Object	Role
Cayley graph $\mathrm{Cay}(G, S)$	symmetric 1-skeleton: vertices and edges
Face system $\mathcal{F}$	choice of cycles filled as faces
Euler characteristic χ	verifies the surface type using $\chi = V - E + F$
Spectral drawing	produces symmetry-preserving coordinates from eigenspaces
Dual graph Γ^*	records adjacency among the chosen faces

Final message

To construct surfaces from Cayley graphs, we must keep two kinds of data separate: the symmetric graph skeleton and the chosen face system.